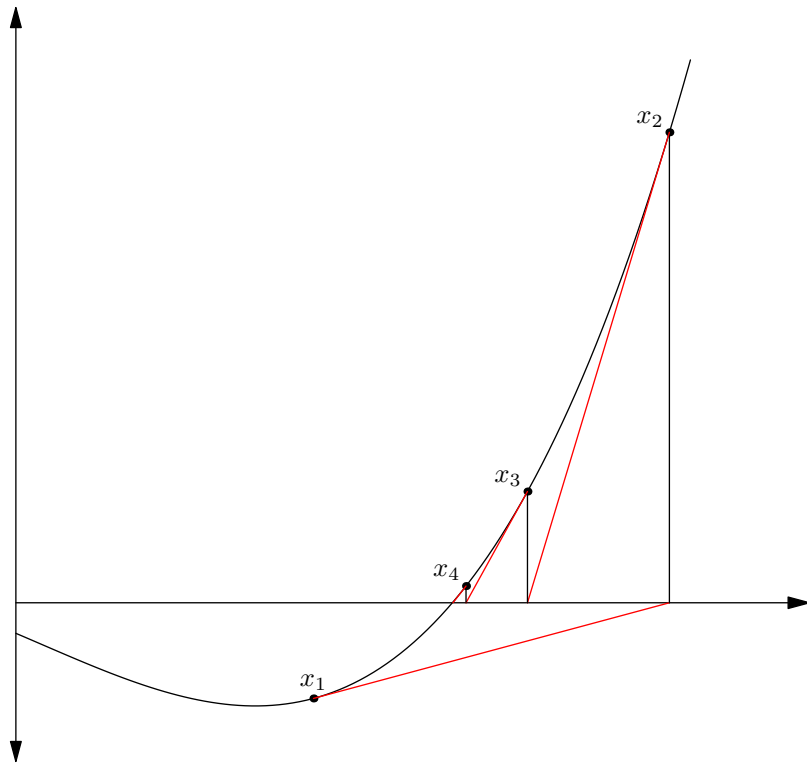


Hensel's Lemma

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§1 Newton's Method

Newton's Method is a very useful method for generating better and better approximations for the roots of a differentiable function, given some initial good approximation. In particular, suppose we want to approximate a root of the differentiable function $f(x)$ and suppose x_1 is close to a root of $f(x)$. Then, the tangent line to $f(x)$ at x_1 is a relatively good approximation to $f(x)$ around x_1 , so the x value where the tangent line intersects the x -axis should be relatively close to the root of $f(x)$; hopefully closer than x_1 at least. Pictorially, approximating a root with Newton's Method looks something like this:



If we want to find arbitrarily good approximations using Newton's Method, we would probably like to iterate it multiple times, so we should write down a formula for calculating the next best approximation given by Newton's Method given one approximation. Let's say that x_n is our current best approximation. Then, the tangent line to $f(x)$ at x_n is $y = f(x_n) + f'(x_n)(x - x_n)$. Solving for $y = 0$, we have

$$xf'(x_n) = x_nf'(x_n) - f(x_n)$$

$$x = x_n - \frac{f(x_n)}{f'(x_n)}$$

We might like to write this as $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ instead to make it clearer that we are getting better and better approximation as we iterate Newton's Method. Another helpful way to think about Newton's Method is through Taylor Series. In particular, suppose that x_n is our approximation that we want to refine using Newton's Method. Then, we can express $f(x)$ as a Taylor series for values around x_n as

$$f(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{f''(x_n)}{2!}(x - x_n)^2 + \frac{f'''(x_n)}{3!}(x - x_n)^3 + \dots$$

Suppose α is the actual root of $f(x)$. Then, $f(\alpha) = 0$. Instead of considering the above expansion as an infinite series, we will disregard all quadratic and higher terms and turn this equality into an approximation as follows:

$$\begin{aligned} f(\alpha) = 0 &\approx f(x_n) + \alpha f'(x_n) - x_n f'(x_n) \\ \implies x_n f'(x_n) - f(x_n) &\approx \alpha f'(x_n) \\ \implies \alpha &\approx x_n - \frac{f(x_n)}{f'(x_n)} \end{aligned}$$

Thus, we let x_{n+1} be this new approximation for α , and we get the same formula which we got before.

§2 Hensel's Lemma

Suppose we are trying to find the roots to the polynomial $f(x) = x^2 - 2 \pmod{7}$. $x = 4$ happens to be a root to this since $4^2 - 2 \equiv 14 \equiv 0 \pmod{7}$. Next, we might want to find a root $\pmod{7^2 = 49}$. If x is a root to $f \pmod{49}$, it certainly is also a root $\pmod{7}$, since all multiples of 49 are multiples of 7. Thus, we can write x in the form $4 + 7k$ where $0 \leq k < 7$. Thus, we want to solve

$$\begin{aligned} (4 + 7k)^2 - 2 &\equiv 0 \pmod{49} \\ \iff 16 + 56k + 49k^2 - 2 &\equiv 0 \pmod{49} \\ \iff 7k + 14 &\equiv 0 \pmod{49} \\ \iff 7k &\equiv 35 \pmod{49} \\ \iff k &\equiv 5 \pmod{7}. \end{aligned}$$

Thus, we take $k = 5$ to see that $4 + 7 \cdot 5$ is a root to $f \pmod{49}$. In fact, we keep on going like this and find roots to $f \pmod{7^m}$ for higher and higher powers of 7, but it is not entirely obvious that this will work. Let's now consider a general polynomial $f(x) = a_m x^m + a_{m-1} x^{m-1} + \dots + a_1 x + a_0$ and a prime p . Suppose $x = x_1$ is a root to $f(x) \pmod{p}$. Then, any root of $f(x) \pmod{p^2}$ must be of the form $x_1 + pk$ for some $0 \leq k < p$. Then, we need

$$\begin{aligned} 0 &\equiv f(x_1 + pk) \pmod{p^2} \\ \iff 0 &\equiv a_m(x_1 + pk)^m + a_{m-1}(x_1 + pk)^{m-1} + \dots + a_1(x_1 + pk) + a_0 \pmod{p^2}. \end{aligned}$$

Before we expand everything out in this equation, notice that any terms which include a factor of p^2 are irrelevant $\pmod{p^2}$, so we need

$$a_m x_1^m + p k a_m (m x_1^{m-1}) + a_{m-1} x_1^{m-1} + p k a_{m-1} ((m-1) x_1^{m-2}) + \dots + a_1 x_1 + a_1 p k + a_0 \equiv 0 \pmod{p^2}.$$

The expansion of $f(x_1)$ is obviously sitting here, and the rest of the terms resemble the derivative of $f(x)$. In fact, if we just let $f'(x)$ be defined based on the power rule from calculus (and not necessarily the limit definition), we see that

$$f(x_1) + pkf'(x_1) \equiv 0 \pmod{p^2}.$$

Recall that we are trying to solve for $x_1 + pk$, which would be our root to $f(x) \pmod{p^2}$. We can do this by dividing by $f'(x_1)$ first, provided that it is not $0 \pmod{p}$, because in general, numbers which are $0 \pmod{p}$ have no inverse $\pmod{p}$ or any power of p . However, if we do assume that $f'(x_1) \not\equiv 0 \pmod{p}$, we have

$$\begin{aligned} \frac{f(x_1)}{f'(x_1)} + pk &\equiv 0 \pmod{p^2} \\ \implies x_1 + pk &\equiv x_1 - \frac{f(x_1)}{f'(x_1)} \pmod{p^2}. \end{aligned}$$

In fact, the above formula is the same as the one we derived for Newton's Method! Now we might as well let $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$. We could obviously keep on going like this, calculating roots of f for higher powers of p . The only thing we need to check is that $f'(x_n)$ will never be $0 \pmod{p}$, given that $f'(x_1) \not\equiv 0 \pmod{p}$. This is true because each of the x_n 's are congruent to $x_1 \pmod{p}$, so because f' is a polynomial, $f'(x_n) \equiv f'(x_1) \not\equiv 0 \pmod{p}$ (write out f' as a polynomial if you're not convinced of this). In general, we can write Hensel's Lemma in the following way:

Theorem 2.1 (Hensel's Lemma)

Let $f(x)$ be a polynomial, p a prime, and n a positive integer. If x_n is a root of $f(x) \pmod{p^n}$ (i.e. $f(x_n) \equiv 0 \pmod{p^n}$) and $f'(x_n) \not\equiv 0 \pmod{p}$, then x_n can be "lifted" to a root $x_{n+1} \pmod{p^{n+1}}$ given by

$$x_{n+1} \equiv x_n - \frac{f(x_n)}{f'(x_n)} \pmod{p^{n+1}}.$$

In other words, $f(x_{n+1}) \equiv 0 \pmod{p^{n+1}}$.

§3 Connection to p -adics

The resemblance of Theorem 2.1 to the formula in Newton's Method is not a coincidence. Recall that Newton's Method generates better and better approximations to the roots of some function in the real numbers. Hensel's Lemma essentially does this same thing in the p -adic integers.

One can think of some p -adic integer α as storing information about $\alpha \pmod{p^n}$ for every single power of p at once. In particular, p -adic integers are usually written as infinite strings of digits going to the left, where each digit is between 0 and $p - 1$ inclusive. For example,

$$\alpha = \overline{\dots 1234560345}_7$$

is an example of a 7-adic integer (note that the notation used above is not standard and will probably be different in other sources). If we wanted to read off the value of $\alpha \pmod{7}$, it would just be the last digit, 5. Similarly, the value of $\alpha \pmod{49}$ is the last two digits, which have a value of a 45, but this is in base 7, so in base 10, $\alpha \equiv 7 \cdot 4 + 5 \equiv 33 \pmod{49}$. As we read off more digits from the right, we can find the value of $\alpha \pmod{p^n}$ for higher powers of p .

Of course, all regular integers can be written as p -adic integers. For example, the base-10 number 97, which is equal to 166_7 (its base-7 representation), can naturally be interpreted as the 7-adic integer

$$\overline{\dots 00000266}_7$$

where there is an infinite string of 0's going to the left. The p -adics obviously contain way more numbers (including rational and even irrational numbers!) since they don't necessarily need to include an infinite string of 0's going to the left.

Now we will discuss a notion of distance for p -adics. Unlike in the real numbers, we can't compare how close two numbers by looking at the leftmost digit, because there may not be any leftmost digit. Instead, it makes more sense to consider two p -adic integers "close" if they agree mod p , even closer if they agree mod p^2 , even closer still if they agree mod p^3 , and so on. Thus, we define the **p -adic absolute value** $|\cdot|_p$ by

$$|\alpha|_p = \frac{1}{p^k}$$

if the last k digits of α as a p -adic number are 0 (and the $k+1$ st isn't). In other words, $\alpha \equiv 0 \pmod{p^j}$ for all $1 \leq j \leq k$. Then, we define the **p -adic distance** between two p -adic integers x and y by $|x - y|$, which is just the number of digits that x and y agree upon, starting from the right.

Under this way of defining distance in the p -adic integers, we can see Hensel's Lemma as an application of Newton's Method for p -adic integers. In particular, if we have a polynomial $f(x)$ and an integer x_1 such that $f(x_1) \equiv 0 \pmod{p}$, Hensel's Lemma will give us an integer x_2 such that $f(x_2) \equiv 0 \pmod{p^2}$. If we think about f as a polynomial over the p -adics which has some root α in the p -adics, x_1 is like our initial approximation. Assuming $x_1 \equiv \alpha \pmod{p}$ (we are assuming this because f could have more roots which are not congruent to x_1), then $|x_1 - \alpha| \leq \frac{1}{p}$ because they agree to at least one digit, so x_1 and α are relatively close. However, once we generate the next approximation x_2 , we have that $|x_2 - \alpha| \leq \frac{1}{p^2}$, so we are getting closer to the true value of α . So, Hensel's Lemma really is just an analogue of Newton's Method.

§4 Examples

In a previous section, we showed the example of computing the square root of 2 mod 7 and mod higher powers of 7. This can be rephrased as trying to find the square root of 2 in the 7-adics. We can of course compute more and more digits of the square root of 2 since the derivative of $x^2 - 2$, which is $2x$, is nonzero mod 2 when evaluated at the square root of 2 (it is congruent to $4 \cdot 2 \equiv 1 \pmod{7}$). In fact, by computing more digits, we have the following approximation of the square root of 2:

$$\sqrt{2} = \overline{\dots 55400245450454}_7.$$

Unfortunately, there are some complications if we try to compute 2-adic square roots. For example, suppose we wanted to find the 2-adic square root of 17, so we are trying to find roots to $x^2 - 17$. This polynomial does have roots mod 2, 4, and even 8, but its derivative, which is $2x$, is 0 mod 2 when evaluated at any of these roots. This means we can't use Hensel's Lemma directly to calculate the full 2-adic expansion of $\sqrt{17}$. We can fix this by instead calculating $\frac{1+\sqrt{17}}{2}$, which you can check satisfies the polynomial $4x^2 - 4x - 16$. The problem here is that this polynomial is always divisible by 4, so at least mod 4, any number will satisfy it. However, if we want to find a root mod higher powers of 2, we can divide out by this factor of 4 and find a root mod a slightly lower power of 2. For example, if we want to find a root mod 8, we really just need to find a root of $x^2 - x - 4 \pmod{2}$, and one of those roots is $x = 1$. To find a root mod 16, we really need to solve $x^2 - x - 4 \pmod{4}$, and

we can lift our root mod 2 using Hensel's Lemma because we know that the derivative is nonzero now! So, theoretically, we should be able to calculate $\frac{1+\sqrt{17}}{2}$ in the 2-adics, and by multiplying that number by 2 and subtracting 1, we will find $\sqrt{17}$.

Exercise 4.1. Show that in fact, any number which is 1 mod 8 has a square root in the 2-adics.

§5 Further Considerations

The one main obstacle for using Hensel's Lemma is having $f'(x_n) \equiv 0 \pmod{p}$ where x_n is one of your approximate roots to some polynomial f . One way this could happen is if f has a repeated root at x_n mod some power of p . In particular, suppose that $f(x)$, when factored as a polynomial mod p^n , has the root x_n twice. In particular, $f(x) \equiv (x - x_n)^2 g(x)$ for some polynomial g . Then, when we differentiate f using the product rule, we get

$$f'(x) \equiv (x - x_n)^2 g'(x) + 2(x - x_n)g(x) \equiv (x - x_n)((x - x_n)g'(x) + 2g(x)) \pmod{p^n}$$

so $f'(x)$ actually has $x - x_n$ as a root, so $f'(x_n) \equiv 0 \pmod{p^n}$. Intuitively, this means that there are at least two roots of f mod p^n , and the formula given by Hensel's Lemma doesn't know which one to converge to. $f(x_n) \equiv 0 \pmod{p}$ could also indicate there aren't actually any p -adic roots of f and only roots mod a couple of powers of p . For example, 5 is a perfect square mod 2 and mod 4 because $1^2 \equiv 5 \pmod{2}$ and $1^2 \equiv 5 \pmod{4}$, but 5 is not a perfect square mod 8. In fact, the only perfect squares mod 8 are 0, 1, and 4. This is due to the fact that the derivative of $x^2 - 5$, which is $2x$, is 0 mod 2 when evaluated at $x = 1$.

Related to this idea of multiple roots, in the examples we showed of Hensel's Lemma, we essentially considered polynomials as only having one root. For example, the only root of $x^2 - 2$ mod 7 that we considered was 4, but 3 is also a root. If you use Hensel's Lemma to lift the root 3 into roots mod higher powers of primes, you will get the other 3-adic square root of 7. This is similar to Newton's Method. In particular, when applying Newton's Method on a real-valued function, the root you converge to depends on which root you are closest to. There are also shortcomings to Newton's Method we didn't discuss. For example, when you have some approximation x_n to the root of $f(x)$, if $f'(x_n) = 0$, then the tangent to line to f at x_n is a horizontal line and will never intersect the x -axis, so you won't be able to generate a better approximation. Newton's Method also exhibits some even more chaotic behavior at times such as cycling between two different values. We won't explore these in depth, but overall, Newton's Method for real-valued functions is more chaotic than Hensel's Lemma, which can usually be used without much effort to both prove the existence of some root as well as calculate its precise value.